

Task 46: Diameter of Binary Tree

Problem Description

Given the root of a binary tree, return the length of the diameter of the tree. The diameter of a binary tree is the length of the longest path between any two nodes in a tree. This path may or may not pass through the root. The length of a path between two nodes is represented by the number of edges between them.

Java Solution

```
class Solution {
    int max = 0;

    public int diameterOfBinaryTree(TreeNode root) {
        maxDepth(root);
        return max;
    }

    private int maxDepth(TreeNode root) {
        if (root == null) return 0;

        int left = maxDepth(root.left);
        int right = maxDepth(root.right);

        max = Math.max(max, left + right);

        return Math.max(left, right) + 1;
    }
}
```